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Quantifying the influence of single particle shape on crushing characteristics of recycled aggregates: Experimental and numerical insights

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ABSTRACT

Exploring the interplay between shapes and crushing characteristics of recycled aggregates (RA) is pivotal for understanding their macro-mechanical behavior in bulk materials. This investigation quantitatively characterizes the shapes of recycled bricks (RB), recycled concrete (RC), and recycled mortar (RM) particles, and assesses their single-particle crushing characteristic. Key focus areas include the analysis of crushing modes, fractal attributes, strength, and energy, supplemented by a comparative study using discrete element method (DEM) simulations to predict outcomes for particles of specific sphericity. Findings reveal that the shapes of RA particles are mainly disk, block, blade and bar, with particle size showing negligible influence on these forms. Crushing behaviors are categorized into penetrating fracture, corner rupture, and overall fragmentation, closely linked to particle sphericity. Notably, an increase in sphericity leads to a reduction in fractal characteristics of RA, refining the gradation distribution from broad to narrow spectrums. An exponential relationship is identified between the crushing strength of RA and particle sphericity, a pattern also mirrored in the correlation between crushing energy and sphericity, thereby underscoring the significant influence of particle shape. These relationships facilitate predictions regarding the crushing strength and energy based on sphericity. While numerical simulations broadly align with experimental data, discrepancies in the crushing energy of highly spheric particles necessitate adjustments, resulting in a modified predictive equation for crushing energy. This work enriches the understanding of RA crushing behavior, offering invaluable insights for predicting the single-particle strength and deformation characteristics across different sphericities, thus advancing material engineering and recycling practices.

1. Introduction

With the rapid development of highway construction in China, the focus has shifted from building new roads to expanding and upgrading existing ones, consequently generating a significant amount of construction and demolition waste [1,2]. Among them, the most representative are bricks, concrete blocks and mortar, accounting for more than 95 % of the total construction and demolition waste [3]. Reusing recycled bricks (RB), recycled concrete (RB), and recycled mortar (RM) processed from construction and demolition waste in infrastructure projects is an effective alternative to traditional natural aggregates. This approach reduces the construction carbon footprint and achieves additional economic benefits [4,5]. However, compared to natural gravel aggregates, recycled aggregates (RA) exhibit notable differences in particle strength and a propensity for crushing [6]. Particle crushing directly affects gradation, which in turn influences the overall strength, deformation, and permeability characteristics of the materials [7,8]. RA particles vary widely in size and shape, making their crushing process complex and influenced by numerous factors, among which particle shape is a significant determinant [9]. Given that a single particle represents the smallest fundamental unit of RA, its crushing characteristics are crucial for the aggregate's overall macro-mechanical performance. Thus, elucidating the crushing and strength characteristics of RA with varying particle shapes is particularly essential for their practical application in the construction of highway.

Current research on single-particle crushing characteristics primarily focuses on brittle geomaterial particles, analyzing their crushing modes, post-crushing particle size distribution, crushing strength, and influencing factors. Under external load, particles exhibit different crushing modes. Tu et al. [10] categorized crushing modes into surface grinding, local fracture, complete fracture, and complete breakage based on the size variation of resulting fragments. Wang et al. [11] conducted

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single-particle crushing tests on Leighton Buzzard sand and completely weathered granite particles, capturing their crushing processes and classifying the crushing modes into splitting, explosive, explosion-splitting, and chipping. Furthermore, numerous scholars have successfully applied fractal theory to analyze the particle size distribution following crushing [12,13,14], providing an effective means to describe the gradation curve distribution of post-crushed geomaterial particles. Crucially, a quantitative mathematical model has been established to link particle crushing strength with the gradation of particle sizes, thereby describing the process of gradation change caused by particle crushing. Zhou et al. [15] conducted numerical simulations of single-particle crushing tests on five groups of rockfill particles with different sizes, revealing a clear size effect on crushing strength. To study the strength of coarse particles with larger diameters, Frossard et al. [16] proposed a method based on size effect to estimate the size of large rockfill particles reasonably. Meng et al. [17] simulated the single-particle crushing process using the discrete element method, extending the approach to larger particle sizes to provide a reference for obtaining the strength of coarse-grained soil with large diameters. Zhao et al. [18] performed in-situ compression tests on sand and granite particles to study their crushing mechanisms, finding that particle morphology and initial microstructure are the primary factors determining crushing modes. Numerous studies [19,20,21,22] have systematically investigated the factors influencing particle crushing, demonstrating that it is affected by multiple factors, including loading conditions, intrinsic particle strength, particle size, and shapes. Among these, particle shapes play a significant role in determining the crushing modes and strength.

Thus, while existing studies have produced considerable findings on the crushing characteristics of brittle geomaterial particles, the diversity of geomaterial particle shapes in practical use presents challenges. On one hand, current research has not delved deeply into the relationship between particle shape and crushing characteristics, and on the other hand, it is difficult to study their macro-mechanical and deformation characteristics through indoor experiments. Moreover, the focus of the above studies has been mainly on the crushing characteristics of traditional natural brittle materials, leaving a research gap for new types of materials such as those derived from construction and demolition waste, which hampers the resource reuse of these materials. To address this, the present study first quantitatively characterizes the particle shape of RA (RB, RC, and RM), and then conducts single-particle crushing tests on RA with different particle shapes (sphericity), investigating their crushing modes, fractal characteristics, crushing strength, and changes in crushing energy. Using the discrete element numerical method, the crushing process of single particles with different sphericities is simulated and compared with experimental data and prediction results to verify the reliability of the numerical program. Finally, the program is extended to specific particle sphericities to analyze their crushing strength and energy, providing a theoretical basis for predicting the strength and deformation of RA with different sphericity.

2. Quantitative characterization of RA single particle shapes

2.1. RA samples and single particle shape indicators

The materials selected for the experiment consist of recycled aggregates derived from the crushing of building demolition debris, sourced from an urban village in Wenling City, Zhejiang Province, China. These are primarily composed of RB, RC, and RM, accounting for over 95 % of the material, as shown in Fig. 1. The characterization of irregular particle shapes generally encompasses three levels [22,23]: contour shape, concave-convexity, and surface texture. For large and irregular particles such as RA, their shape characteristics can be described using two levels: contour shape and concave-convexity. In this experiment, a precision electronic vernier caliper (accuracy of 0.1 mm) was used to accurately measure the length of each RA particle in three mutually perpendicular directions: x, y, and z. The maximum value is denoted as L, the intermediate value as I, and the minimum value as S, as shown in Fig. 1(e).

The four common indicators were chosen to describe the shaped differences of RA particles: elongation index (EI), flatness index (FI), Domokos coefficient (S_f), and sphericity (Φ), calculated as per Eqs. (1), (2), (3), and (4) respectively. To generalize the results, 45 particles of three different recycled aggregates across various particle sizes (4.75–9.5 mm, 9.5–19 mm, and 19–37.5 mm) were measured, totaling 405 particles (45×3×3=405). To characterize the dispersion of the measured data of RA particle size and shape parameters, the sample variance s^2 was calculated in Table 1. The s^2 is an unbiased estimate of the overall variance, which is suitable for the case of only sample data. According to the 405 samples provided in this study, the s^2 calculated by grouping was listed in Table 1. This result reflects the change of RA particle size and shape parameter data points around the mean value. It can be seen that with the increase of particle size, the discrete degree of

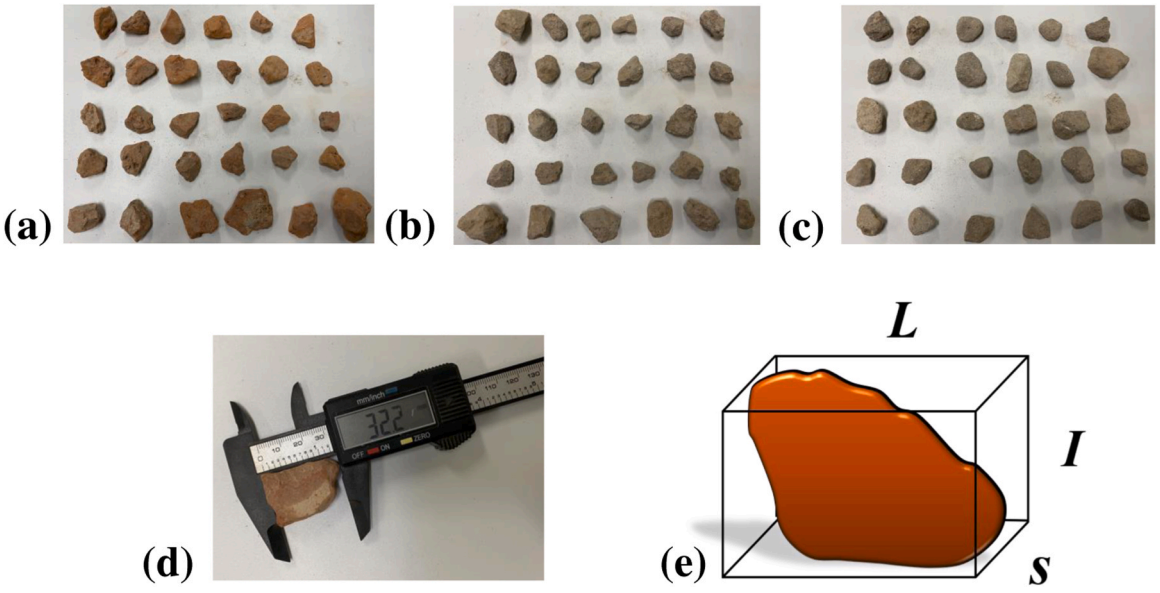


Fig. 1. RA particle samples and particle size measurement. (a) RB samples; (b) RC samples; (c) RM samples; (d) A precision electronic vernier caliper; (e) Quantitative illustration of RA particle size.

Table 1
Statistical analysis of average shaped indicators of RA with different particle sizes.

Component categories	Particle size range(mm)	L(mm)	I(mm)	S(mm)	EI	FI	S_f	Φ
RB	4.75–9.5	14.07 (s^2=0.064)	8.80 (s^2=0.026)	5.42 (s^2=0.016)	0.64 (s^2=0.019)	0.64 (s^2=0.033)	3.95 (s^2=0.664)	0.63 (s^2=0.013)
	9.5–19	23.78 (s^2=0.229)	16.76 (s^2=0.084)	10.42 (s^2=0.092)	0.73 (s^2=0.024)	0.63 (s^2=0.038)	3.74 (s^2=0.340)	0.68 (s^2=0.010)
	19–37.5	35.80 (s^2=0.546)	26.33 (s^2=0.359)	16.09 (s^2=0.335)	0.75 (s^2=0.020)	0.63 (s^2=0.052)	3.79 (s^2=0.505)	0.69 (s^2=0.016)
RC	4.75–9.5	13.0 (s^2=0.057)	8.62 (s^2=0.019)	5.64 (s^2=0.011)	0.68 (s^2=0.021)	0.67 (s^2=0.026)	3.69 (s^2=0.403)	0.66 (s^2=0.010)
	9.5–19	25.24 (s^2=0.176)	17.78 (s^2=0.067)	11.98 (s^2=0.059)	0.72 (s^2=0.017)	0.69 (s^2=0.035)	3.56 (s^2=0.247)	0.70 (s^2=0.008)
	19–37.5	36.07 (s^2=0.538)	25.09 (s^2=0.241)	16.76 (s^2=0.269)	0.71 (s^2=0.025)	0.68 (s^2=0.034)	3.65 (s^2=0.295)	0.69 (s^2=0.013)
RM	4.75–9.5	12.38 (s^2=0.040)	8.56 (s^2=0.028)	5.51 (s^2=0.012)	0.70 (s^2=0.021)	0.66 (s^2=0.015)	3.61 (s^2=0.170)	0.68 (s^2=0.009)
	9.5–19	23.33 (s^2=0.102)	16.67 (s^2=0.102)	10.20 (s^2=0.103)	0.73 (s^2=0.024)	0.63 (s^2=0.043)	3.77 (s^2=0.362)	0.67 (s^2=0.010)
	19–37.5	34.13 (s^2=0.346)	26.18 (s^2=0.356)	15.58 (s^2=0.280)	0.78 (s^2=0.021)	0.62 (s^2=0.044)	3.73 (s^2=0.440)	0.70 (s^2=0.011)

Note: s^2 represents the sample variance.

statistical results of particle size parameters of RA tends to be larger. Except for S_f, the discrete degree of EI, FI and Φ of shape parameters is low, and the statistical data is relatively stable.

$$EI = \frac{I}{L} \tag{1}$$

$$FI = \frac{S}{L} \tag{2}$$

$$S_f = \frac{\left(\frac{1}{S}+\frac{1}{I}+\frac{1}{L}\right)\sqrt{S^2+I^2+L^2}}{\sqrt{3}} \tag{3}$$

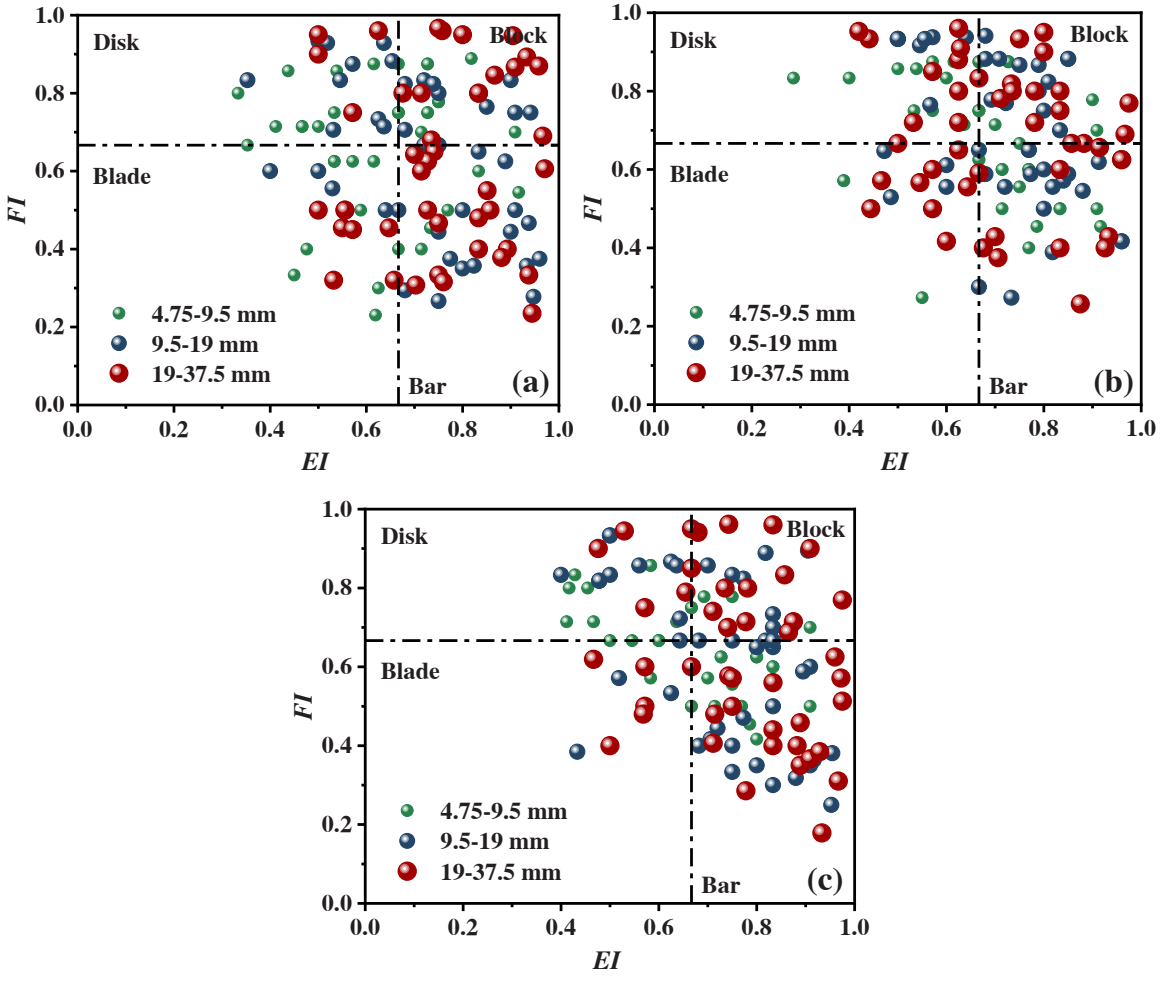


Fig. 2. Classification of particle shape of RA within different particle size ranges. (a) RB; (b) RC; (c) RM.

$$\Phi = \sqrt[3]{\frac{IS}{L^2}} \tag{4}$$

where L represents the longest axis of the particle in millimeters (mm); I denotes the intermediate axis of the particle in millimeters (mm); S signifies the shortest axis of the particle in millimeters (mm).

2.2. Quantitative characterization of single-particle shapes

Based on the basic shape parameters of particles obtained through precise measurements, a bivariate method was employed to classify the shapes of RA particles from RB, RC, and RM. The statistical results are illustrated in Fig. 2. As can be seen from Fig. 2, the shapes of RA particles mainly include disk, block, blade and bar. Among these, the particle shape distribution of RB is the most widespread. The distribution of RC particle shape is relatively concentrated around the point (EI=2/3, FI=2/3). In addition, the particle size of RA has no effect on the particle shape. For example, the particle shape distributions of RB, RC and RM with different particle sizes are mixed, and there is no significant difference.

Fig. 3 presents the frequency distribution of the particle shape parameter Φ and Domokos coefficient S_f for RA. It is evident from Fig. 3 that the description of particle sphericity Φ follows a Gaussian distribution, while the description of Domokos coefficient S_f follows an

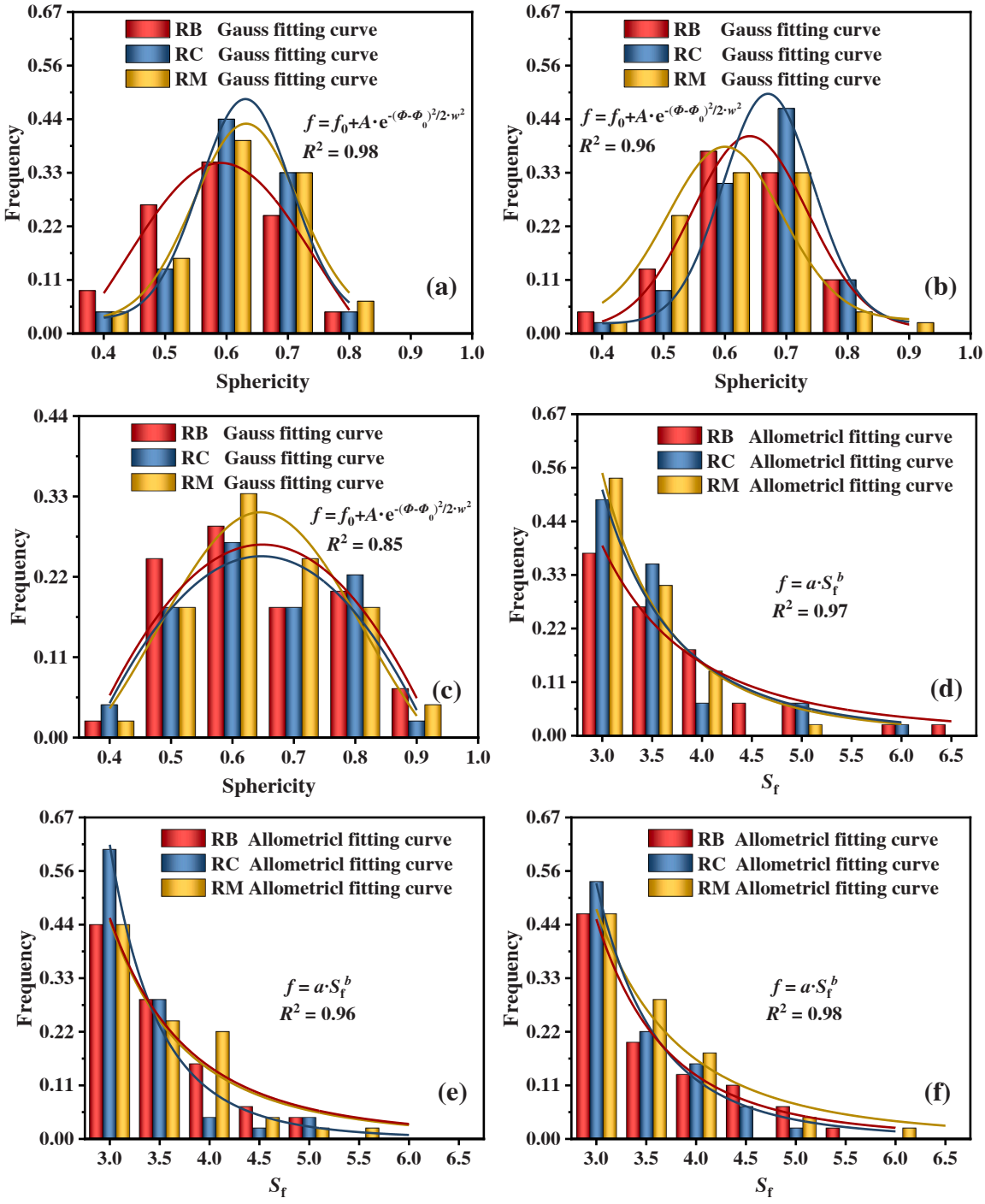


Fig. 3. Frequency distribution map of particle shape parameters. (a)(d) d=4.75–9.5 mm; (b)(e) d=9.5–19 mm; (c)(f) d=19–37.5 mm.

exponential distribution. From Fig. 3(a)(b)(c), it is observed that the distribution of RA particle sphericity Φ ranges between 0.4 and 1.0, with the highest frequency in the Φ=0.5–0.7 region. Moreover, as the particle size increases, the frequency distribution range of the particle sphericity becomes more uniform, which results in a correlation coefficient of the Gaussian distribution of only 0.85 for the particle size d=19–37.5 mm. From Fig. 3(d)(e)(f), it is known that the Domokos coefficient S_f values range between 3.0 and 6.6, but are mainly concentrated between 3.0 and 4.5, accounting for more than 90 % of the distribution. This indicates that the anisotropy of the main axis of the RA particles used in this study is small, meaning that the majority of particle shapes are nearly spherical.

In summary, the flatness of the RA particles used in this study ranges between 0.30 and 0.96, the elongation ratio between 0.29 and 0.98, the Domokos coefficient values between 3.03 and 6.58, and the sphericity between 0.42 and 0.93. Simultaneously, the impact of particle size on particle shape is minimal. Therefore, in the subsequent single-particle crushing tests, particles from the same size range will be selected for testing.

3. Single-particle crushing experiments and discrete element modeling

3.1. Single-particle crushing experiment

Single-particle crushing tests were conducted on three types of RA. The testing apparatus was the MTS universal testing machine in the Geotechnical Laboratory of Chang'an University (Fig. 4(a)), with a maximum load capacity of 100 kN. The testing machine has a maximum load capacity of 100 kN and operates in displacement control mode. This study carried out single-particle crushing tests under different particle shapes for RB, RC, and RM, with the loading method shown in Fig. 4(b). To eliminate the influence of particle size on particle crushing, particles with a particle size range of 19–37.5 mm were selected. The maximum particle size range of RA was selected because the larger particles had a higher crushing probability in the actual engineering particle group. The specific experimental design is presented in Table 2, totaling 270 particles. For each set, 30 tests were conducted to meet the error requirements [24]. Meanwhile, significant differences in the sphericity of RA particle shapes were found in the statistics of Section 2. Therefore, sphericity (Φ) was chosen as the index for dividing the appearance shapes of single particles, and single-particle crushing tests were conducted on particles in three categories: sphericity Φ=0.4–0.6 (Flat spherical), 0.6–0.8 (Subspherical), and 0.8–1.0 (Nearly spherical).

Table 2
Single-particle crushing test schemes with different shapes.

Component categories	Particle size range/ mm	Test the number of particles	Particle sphericity Φ	Axial loading rate/ (mm·min^{-1})
RB	19–37.5	30	0.4–0.6	1.0
		30	0.6–0.8	
		30	0.8–1.0	
RC	19–37.5	30	0.4–0.6	1.0
		30	0.6–0.8	
		30	0.8–1.0	
RM	19–37.5	30	0.4–0.6	1.0
		30	0.6–0.8	
		30	0.8–1.0	

The specific single-particle crushing test process is as follows.

- First, before conducting the single-particle crushing test, carefully place the particle sample at the center of the bottom plate of the universal testing machine to ensure the sample is stable and not tilted (optimal stability). Next, slowly adjust the loading plate of the testing machine, allowing it to gradually descend until it just makes contact with the top surface of the particle sample. Once the loading plate is in full contact with the particle surface, set the loading speed of the testing machine to a constant rate of 1 mm/min. This ensures the acquisition of uniform and continuous force-displacement data. Among them, the loading rate setting method follows the approach used in References [9,22]. Additionally, using a constant loading rate allows for a more focused examination of the influence of particle shape on the crushing characteristics of single particles, which is the primary focus of this study.
- Start the testing machine and begin applying compressive load, continuously monitoring and recording the force-displacement relationship of the particle throughout the compression process (Fig. 4c). High-precision sensors and data acquisition systems are used to ensure the accuracy and reliability of the data. Observe the deformation and crushing process of the particle. When the force no longer increases during loading, it indicates that the particle has reached its crushing limit, at which point the loading is stopped. Record the maximum load value and the corresponding displacement value as key indicators for evaluating the crushing strength of the particle.
- After unloading, carefully remove the crushed particle sample and inspect and document its morphology for further analysis of the crushing mechanism. Export the force-displacement data and

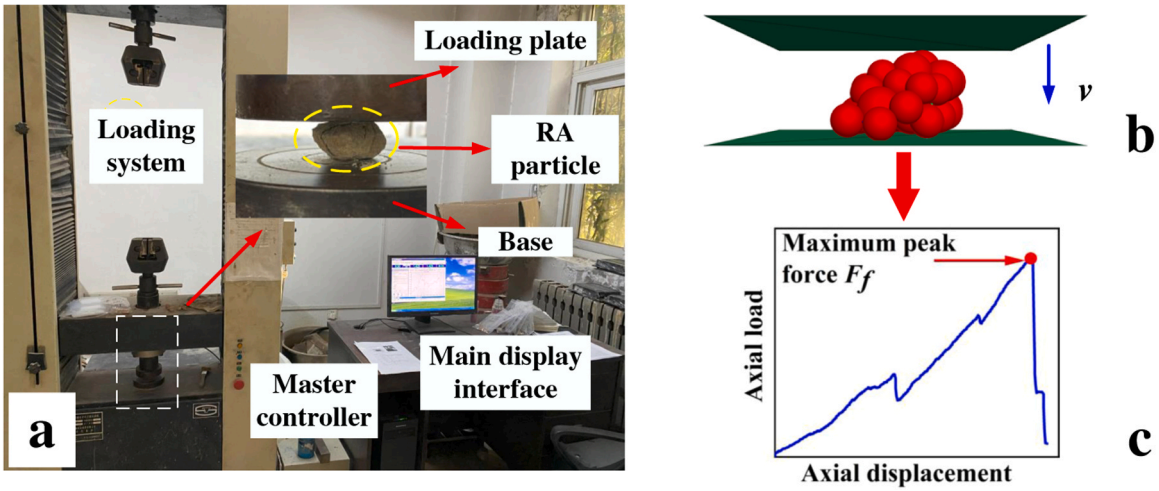


Fig. 4. Experimental equipment, single-particle loading discrete element model and its typical force-displacement curve. (a) MTS universal testing machine; (b) Discrete element simulation of single particle loading; (c) Typical load-displacement curve.

perform the necessary data processing and analysis to plot the final force-displacement curve. By comparing the force-displacement curves of different particles, investigate their crushing strength, crushing models, and energy absorption characteristics. Additionally, combine these findings with observations of the post-crushing morphology to further discuss the mechanical properties and crushing behavior of the RA particle materials.

3.2. Discrete element modeling of single-particle crushing

Using the PFC^{3D} discrete element numerical software, particles of different sizes were bonded into a Clump cluster to simulate a single particle of RA. To ensure the accuracy of the single particle crushing process simulation, the number of sub-particles forming a single RA particle was kept above 1500. For simplicity, an ellipsoid was used to equivalently simulate the single particle model, consistent with the particle shape classification standard in Section 3.1, using sphericity (Φ) as the index to divide the appearance shape of the single particle discrete element model. The single particle and loading process are illustrated in Fig. 5.

Additionally, the sub-particles composing a single particle were bonded using the Bonded Particle Model (BPM) [25], and after particle crushing, they degenerated into a linear contact model. In this study, the mesoscopic parameters of the numerical model were repeatedly calibrated by single-particle crushing test. By adjusting the initial parameters to match the macroscopic characteristics of the RA particles with the macroscopic characteristics measured in the laboratory, this method has strong applicability [26,27]. The initial test parameters in this study mainly refer to Section 3.1 single-particle crushing tests and Literature [9,28,29]. Finally, the key mesoscopic parameters of the numerical simulation of RB, RC and RM were determined by repeated calculation. Specifically, the elastic modulus of RB, RC, and RM are 8.6, 15.2, and 9.1 GPa, respectively, and the density is 2558, 2860, and 2614 kg/m^3, respectively, with a Poisson's ratio of 0.20 for all. In addition, the surface friction coefficient of RA particles used in practical engineering may be different, but this study mainly focuses on the influence of particle shape on the crushing characteristics, so the friction coefficient can be used within a reasonable range. Reference [9,28] shows that the friction coefficient of 0.5 is a reasonable value. Furthermore, referring to the Literature [17], the bonding strength between sub-particles was set to follow a normal distribution. The determination of the mesoscopic parameters for RB, RC, and RM was strictly based on the results of the indoor single-particle crushing tests, with iterative calculations until the mechanical response demonstrated by the numerical simulation was essentially consistent with the trend reflected by the indoor tests. Take an RB particle with d=25.36 mm and sphericity Φ=0.70 as an example, as shown in Fig. 6. When a single particle crushes under the action of the loading plate, it is the bond between the sub-particles inside the single particle that fractures [30]. From Fig. 6, it can be seen that the numerical calculation results of single-particle crushing are essentially in agreement with the experimental test results, with only a 0.74 % difference in the peak force F_f of particle crushing. This indicates that the use of the discrete element numerical method to simulate the single-particle crushing process of RA is feasible.

4. Experimental results and discussion

4.1. Single-particle crushing modes

The single-particle crushing modes of geotechnical bodies can be varied [31], and this paper categorized the results of compressed and crushed RA particles of different shapes (sphericity Φ) based on criteria such as particle size after crushing and cracking method. It was found that the crushing modes of RB, RC, and RM particles can be divided into three typical types: penetrating fracture, corner rupture, and overall fragmentation, as shown in Fig. 7. *Penetrating fracture*: The particle

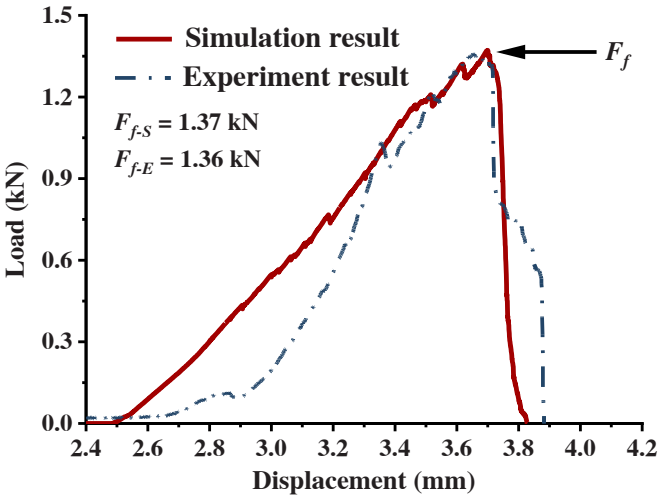


Fig. 6. Comparison between discrete element numerical calculation results and experimental results.

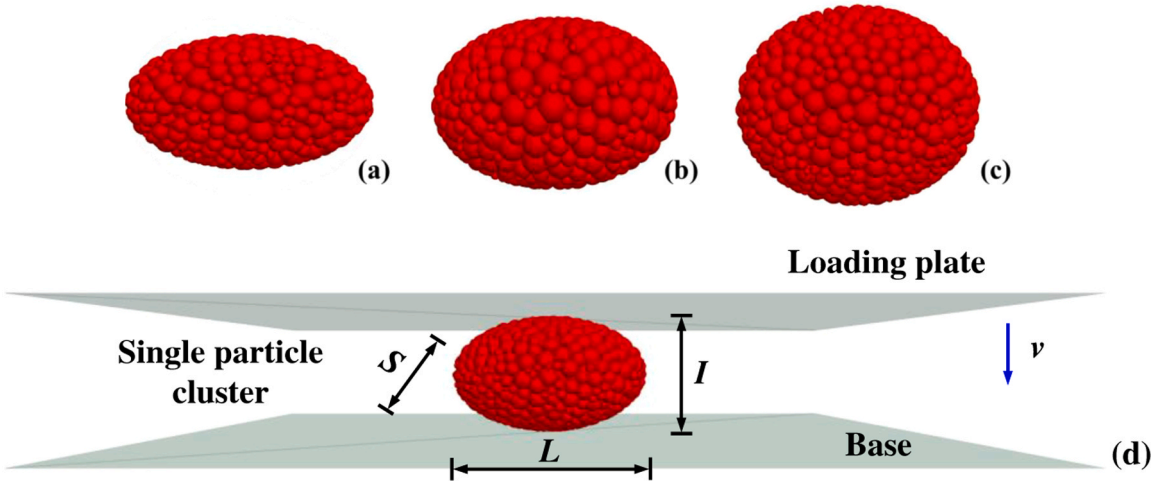


Fig. 5. A typical discrete element model for single particle crushing. (a)(b)(c) Single particle models with different sphericity, and the sphericity Φ=0.50, 0.70, 0.90, respectively; (d) Schematic diagram of single particle loading.

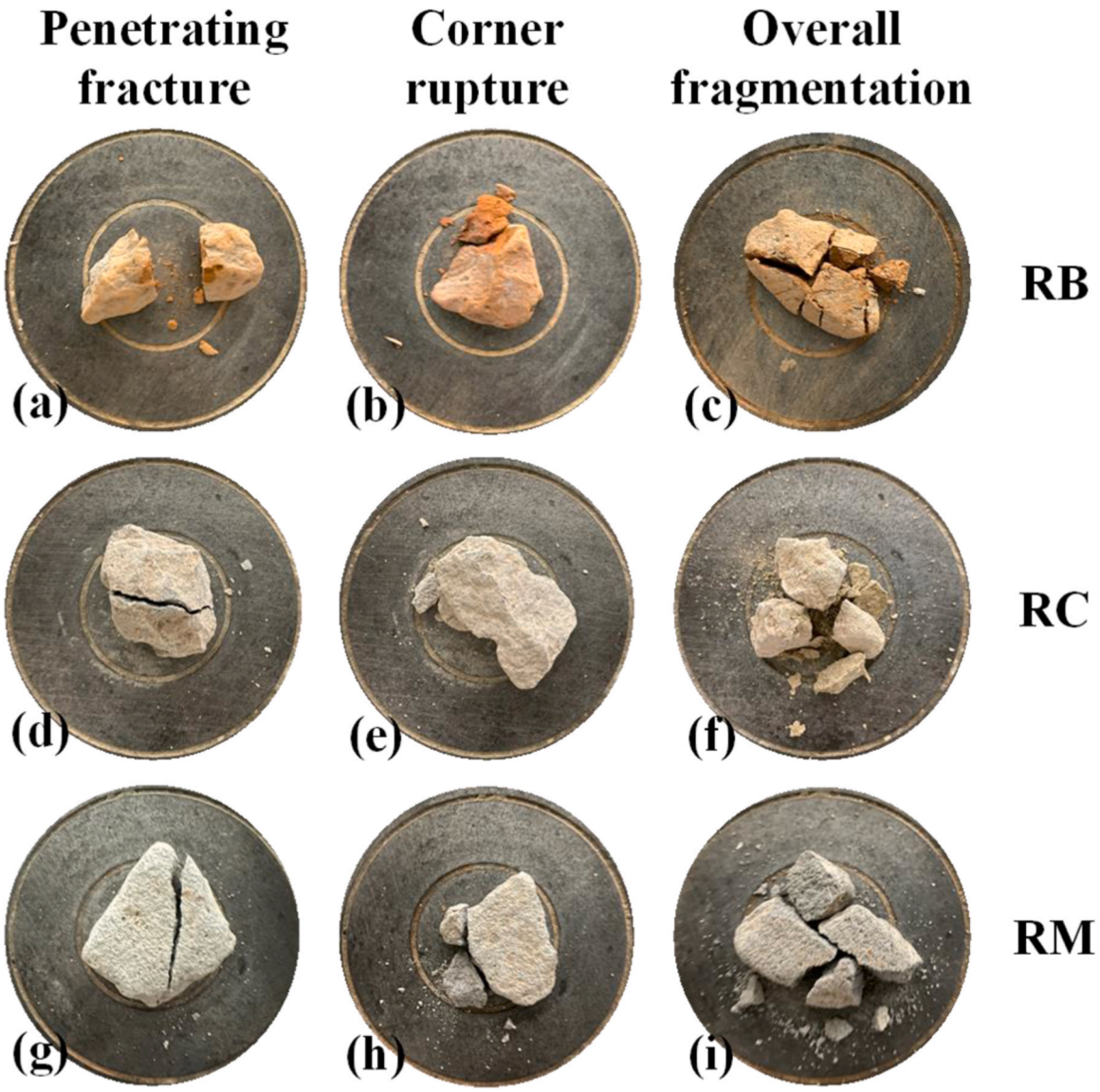


Fig. 7. Typical single-particle crushing modes. (a)(d)(g) Penetrating fracture of RB, RC and RM; (b)(e)(h) Corner rupture of RB, RC and RM; (c)(f)(i) Overall fragmentation of RB, RC and RM.

breaks from the middle after being compressed, generally breaking into two pieces, with a few fragments and relatively neat fracture surfaces, as shown in Fig. 7(a)(d)(g). The proportions of penetrating fracture modes for RB, RC, and RM single particles are 17.78 %, 25.56 %, and 23.33 %, respectively. *Corner rupture*: Most of them break at the corners of the particles. When the fracture occurs, cracks usually form at the corner positions, resulting in the main body breaking into two or three pieces (one super-large piece and some smaller pieces) with some debris. The fracture surfaces is irregular, which is different from the crushing mode of penetrating fracture. This crushing mode indicates that the particles are subjected to large stress concentrations at the corners, resulting in local crushing, as shown in Fig. 7(b)(e)(h). The proportions of corner rupture modes for RB, RC, and RM single particles are 22.22 %, 21.11 %, and 20.00 %, respectively. *Overall fragmentation*: The particles halted the test due to significant damage to the main body, eventually breaking into multiple pieces of similar sizes, accompanied by a substantial amount of debris. The fracture surfaces exhibited extreme irregularity, indicating comprehensive structural damage. The overall fragmentation mode suggests that the particles experienced relatively uniform stress throughout the main body, leading to comprehensive fragmentation, as shown in Fig. 7(c)(f)(i). The proportions of overall fragmentation modes for RB, RC, and RM single particles are 60.00 %, 53.33 %, and 56.67 %, respectively. This indicates that the main mode of single particle crushing for RB, RC, and RM is overall fragmentation. Among them, RB has the highest number of secondary particles after crushing, followed by RM, and RC has the least. This could further predict the amount of particle crushing for RB, RC, and RM in actual engineering applications based on the proportion of different component RA crushing modes.

Furthermore, the shape of the particles also affects the mode of particle crushing to a certain extent. As shown in Fig. 8, a frequency distribution histogram was established for different macroscopic particle shapes (sphericity Φ) and crushing modes. It is evident that the proportion of crushing modes differs among different types of RA particles, but there is a correlation between the single-particle crushing mode of RB, RC, and RM and the particle sphericity Φ. Specifically, the proportions of penetrating fracture and corner rupture modes decrease to some extent with the increase of sphericity Φ, while the proportion of overall fragmentation mode increases with the increase of sphericity Φ. The internal reason is that as the particle sphericity Φ increases, the particles become fuller and the stress distribution outside the particles is more uniform [23], which will reduce the occurrence of penetrating fractures and corner rupture. Meanwhile, according to the above

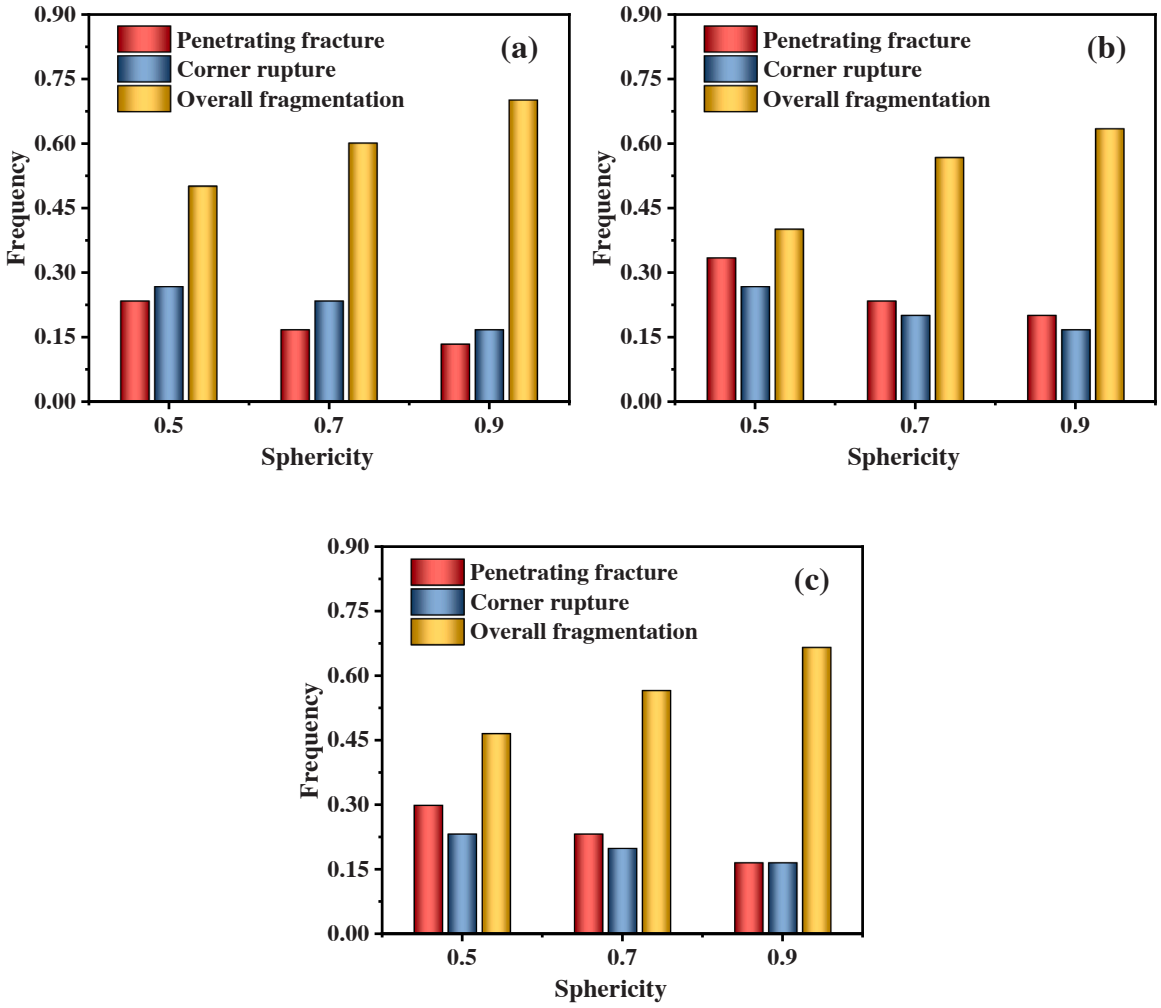


Fig. 8. Frequency distribution histogram of different particle sphericity (Φ) and crushing modes. (a) RB; (b) RC; (c) RM.

analysis, it can be seen that these two crushing modes are usually caused by uneven external stress (stress concentration), which is also consistent with the previous findings [23]. Overall, the sphericity Φ of particles affects the shape, structure, and external stress distribution of the particles, thereby affecting the modes of particle crushing. Further, the crushing mode scenarios for RB, RC, and RM particles of different shapes (sphericity Φ) can provide an important basis for subsequent discrete element simulations of particle crushing involving various shapes.

4.2. Single-particle crushing size fractal characteristics

The process of single-particle crushing essentially leads to changes in the gradation curve of particle sizes. According to fractal theory, the gradation curve of particle sizes follows a fractal distribution [32], and the gradation curve of the particles can be expressed by Eq. (5).

$$M(d) = \left(\frac{d}{d_{\max}}\right)^{3-D} \tag{5}$$

where $M(d)$ represents the mass proportion of particles smaller than a certain particle size; $d_{\max}$ is the largest particle size in the sample; d is the current particle size; and D is the fractal dimension.

After the crushing of single RA particles, detrital grains with different particle sphericities Φ of RB, RC, and RM were collected. These detrital grains were then sieved to determine the particle size distribution post-crushing, as shown in Fig. 9. From Figs. 9(a), 9(b), and 9(c), it can be observed that due to crushing, the gradation curves of RA particles generally shift upwards, and the degree of crushing for RB and RM is greater than that for RC. This indicates that the overall strength of RC is higher than that of RB and RM. Furthermore, under a double logarithmic coordinate system, the fractal dimensions for RB and RM at different initial particle sphericities Φ are greater than that for RC, and the fractal dimension for all three types of RA decreases as particle sphericity increases, indicating a reduction in the degree of crushing. Additionally, as particle sphericity increases, the fractal characteristics of RA weaken, and the distribution of gradation narrows from broad, mainly because RA with larger sphericity are less completely crushed after reaching peak strength in the experiments compared to those with smaller sphericity, resulting in fewer small particles after crushing and thus weaker fractal characteristics.

Subsequently, particles of the same material with three different particle sphericities Φ were mixed together and sieved, and the resulting final gradation curve is shown in Fig. 9(d). In Fig. 9(d), the particle gradation curves of RB, RC and RM in the double logarithmic coordinate system show that the slopes for the multi-sphericity RB, RC, and RM are 0.53, 0.69, and 0.51, respectively. According to Eq. (5), the final fractal dimensions for the multi-sphericity Φ recycled aggregates RB, RC, and RM are 2.47, 2.31, and 2.49, respectively. Additionally, in subsequent experiments on the crushing of RA with multi-sphericity or in discrete element numerical simulations of particle crushing, the gradation of single particle crushing in RA can be determined based on Eq. (5) and the experimentally obtained fractal dimension D.

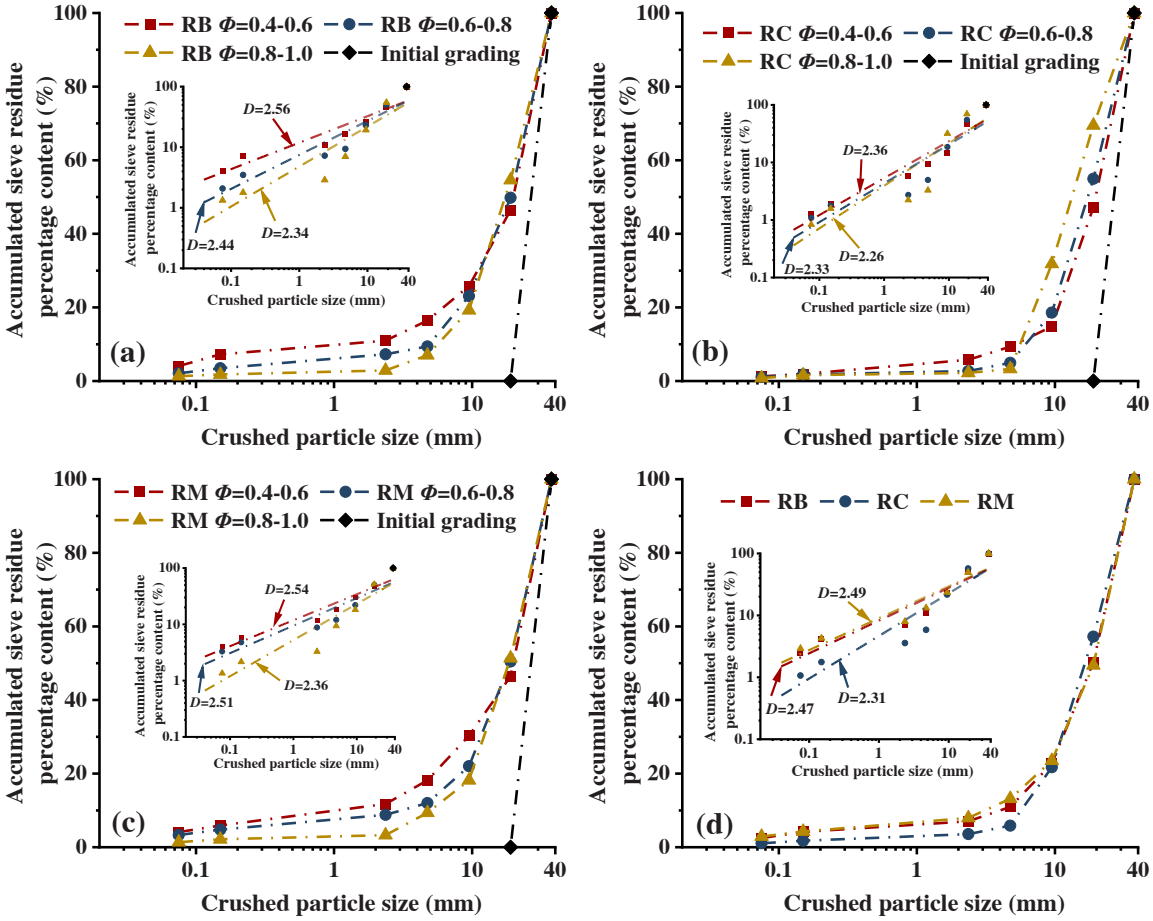


Fig. 9. Single-particle crushing gradation curve and fractal relationship. (a) RB of different particle shapes; (b) RC of different particle shapes; (c) RM of different particle shapes; (d) Particle size distribution and ultimate fractal dimension for the single particle crushing tests.

4.3. Single-particle crushing strength

According to the single-particle strength theory [33], the crushing strength of a single RA particle can be determined using Eq. (6), and the equivalent particle size d can be calculated using Eq. (7).

$$\sigma_f = 0.9\frac{F_f}{d^2} \tag{6}$$

$$d = \sqrt[3]{LIS} \tag{7}$$

where σ_f is the single-particle crushing strength; F_f is the magnitude of the peak particle crushing force obtained in the experiment.

For the crushing of coarse-grained geomaterial particles, the shape of the particle (sphericity Φ) has a significant impact on its crushing strength. However, many studies on the crushing of coarse geomaterial particles have not considered the impact of particle shape [17,34,35]. To

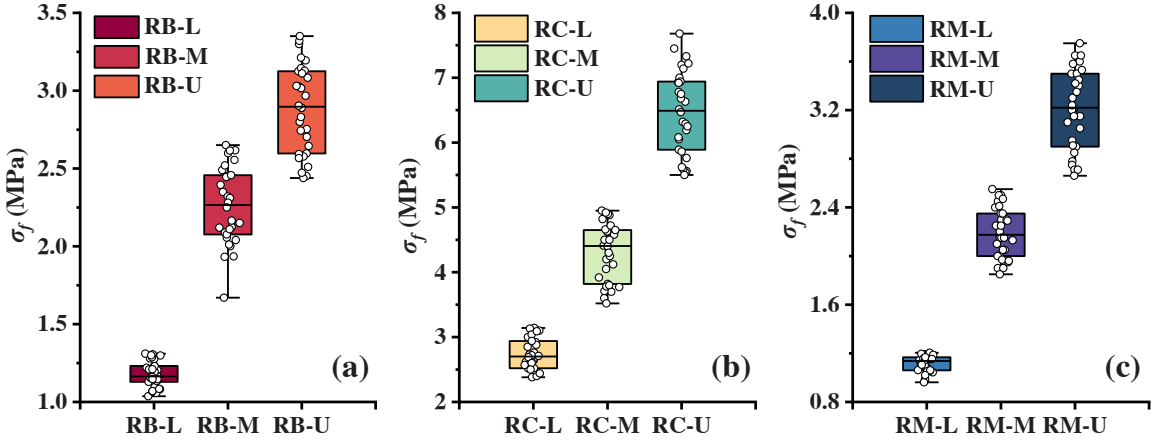


Fig. 10. The crushing strength of RA particles in different particle shape intervals, L, M and U represent Φ = 0.4–0.6, Φ = 0.6–0.8 and Φ = 0.8–1.0, respectively. (a) RB; (b) RC; (c) RM.

enhance the reliability of the test results, this study used the box plot in the statistical method to statistically analyze the 9 sets (270 particles) of crushing strength results of the test RA particles, as shown in Fig. 10. It can be seen from Fig. 10 that the particle crushing strength data of the test are evenly distributed, and the data are all within the upper and lower limits of the box plot, with no abnormal values. In addition, the mean absolute percentage error (MAPE) of each group of test particle crushing strength data is calculated respectively, which is within 5.10 %-9.35 %. Referring to the relevant research [17], it is within a reasonable range and can be accepted. In the follow-up study, the average single-particle peak crushing strength is used to represent the single-particle crushing strength σ_f. The distribution relationship between the single-particle crushing strengths of RB, RC, and RM and particle sphericity is shown in Fig. 11. Through the analysis of their particle sphericity, the relationship between particle sphericity and crushing strength can be expressed in the form of an exponential Eq. (8), where p and q are fitting parameters.

$$\sigma_f = p\Phi^q \tag{8}$$

It can be seen from Fig. 11 that the single particle crushing strength of RC is greater than that of RB and RM for particles of comparable sphericity, which is consistent with the conclusions obtained in Section 4.2. It is found from Fig. 11 that as the particle sphericity increases, the crushing strength of RA also increases. When the sphericities of RB, RC, and RM are 0.531, 0.551, and 0.542, their crushing strengths are 1.17 MPa, 2.74 MPa, and 1.12 MPa, respectively. However, when the particle sphericity increases to 0.856, 0.837, and 0.847, the crushing strengths of RB, RC, and RM increase to 2.88 MPa, 6.47 MPa, and 3.20 MPa, respectively, representing increases of 146.15 %, 136.13 %, and 185.71 %. When particles have a comparable size, greater sphericity means the force is more evenly distributed across the particle, and a greater force is required to crush the particle [23]. Furthermore, it can be seen from Fig. 11 that using Eq. (8) to fit the relationship between single-particle sphericity and crushing strength results in a curve that fits well with the experimental results. The correlation coefficients R^2 of the relationship curves between single-particle sphericity and crushing strength for RB, RC, and RM are 0.90, 0.99, and 0.98, respectively, indicating a significant effect of particle shape.

4.4. Single-particle crushing energy

The crushing energy of a single particle is generally not directly measurable through indoor experiments. However, it can be calculated based on the single particle crushing force-displacement curve and the definition of work [36], as shown in Fig. 12. The crushing energy of a single particle can be expressed as the integral function of the crushing force concerning displacement, as shown in Eq. (9).

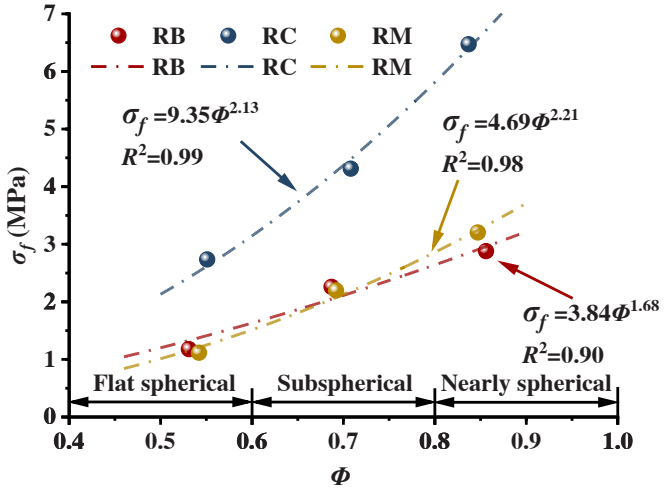


Fig. 11. The relationship between single-particle shape and crushing strength.

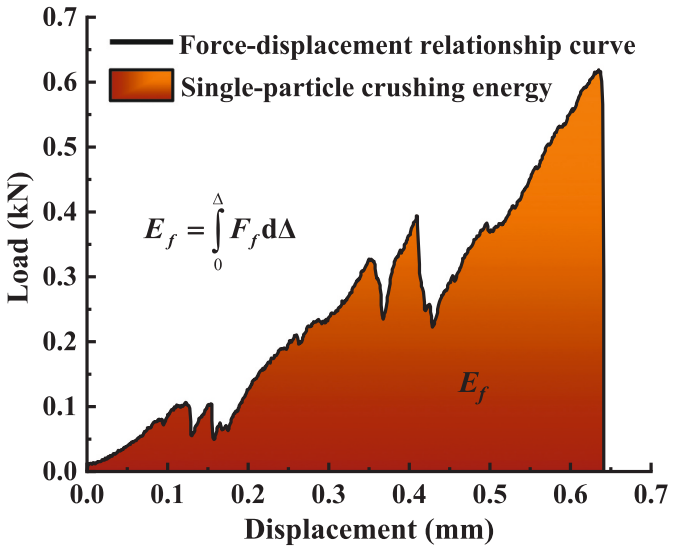


Fig. 12. Typical load-displacement relationship curve and work calculation diagram of single particle crushing.

$$E_f = \int_0^{\Delta} F_f d\Delta \tag{9}$$

where E_f is the single-particle crushing energy, and Δ is the displacement.

Similar to the single-particle crushing strength, to further evaluate the reliability of the single-particle crushing energy data calculated from the test results, the box plot was used to statistically analyze the 9 sets (270 particles) of crushing energy results of the single particle of the recycled aggregate, as shown in Fig. 13. It can be seen from Fig. 13 that the overall distribution of particle crushing energy data is relatively uniform, and there is no abnormal value. The MAPE of each group of particle crushing energy data is calculated, which is within 1.08 %-10.1 %, and MAPE is within a reasonable range [17]. Besides, the particle crushing energy is not the same even if the particles with the same material properties. Therefore, the average single-particle crushing energy under the same particle shape (sphericity Φ) is used to replace the single-particle crushing energy E_f. The distribution relationship between particle crushing energy and sphericity is shown in Fig. 14, which can be expressed in the form of an exponential function as shown in Eq. (10), where m and n are fitting parameters.

$$E_f = m\Phi^n \tag{10}$$

It is evident from Fig. 14 that under the same particle sphericity Φ conditions, the single-particle crushing energy of RC is higher than that of RB and RM. When d=19–37.5 mm, the single-particle crushing energy of RC is 97.98 % and 64.22 % higher than that of RB and RM, respectively. This is mainly because the main components forming the RA are different. Generally, the strength of concrete blocks is higher than that of bricks and mortar, so even RC, RB, and RM particles of the same size and sphericity Φ contain more energy in RC particles, and thus, more energy is released during crushing. Additionally, it can also be found from Fig. 14 that under the same particle size, the closer the particle shape is to a sphere, the greater the crushing energy of a single RA particle. When Φ ranges from 0.4 to 0.6, the single-particle crushing energies of RB, RC, and RM are 349.05 mJ, 912.42 mJ, and 316.62 mJ, respectively. However, when Φ ranges from 0.8 to 1.0, the single-particle crushing energies of RB, RC, and RM are 1288.89 mJ, 1598.63 mJ, and 1381.18 mJ, respectively, showing increases of 269.26 %, 75.21 %, and 336.23 %.

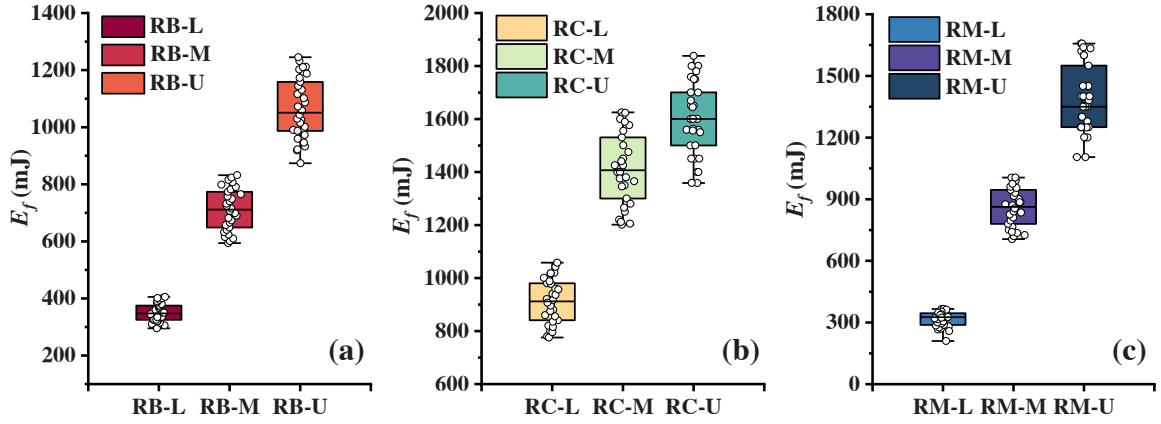


Fig. 13. The crushing energy of RA particles in different particle shape intervals, L, M and U represent Φ = 0.4–0.6, Φ = 0.6–0.8 and Φ = 0.8–1.0, respectively. (a) RB; (b) RC; (c) RM.

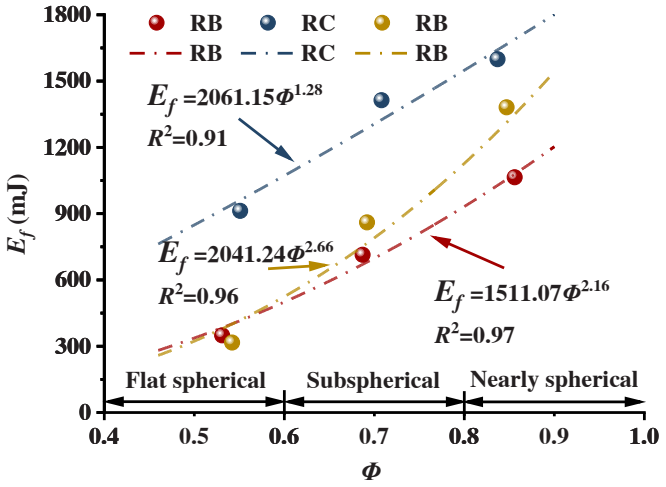


Fig. 14. The relationship between single-particle shape and crushing energy.

In summary, similar to particle crushing strength, the single-particle crushing energy also follows an exponential function relationship with particle sphericity, and it fits well (correlation coefficient (R^2>0.90). Therefore, the single-particle crushing strength and crushing energy of RA of any sphericity can be predicted through particle Φ using Eqs. (8) and (10).

5. Numerical results and discussion

By analyzing single-particle crushing experiments of RA with different shapes (sphericity Φ), the average crushing strength and energy can be predicted using Eqs. (8) and (10). The PFC^{3D} discrete element numerical simulation is employed to evaluate the crushing strength and energy of single particles with sphericities Φ=0.4–0.6, 0.6–0.8, and 0.8–1.0, all within a similar particle size range (19–37.5 mm). These simulations are then compared with experiments to validate the parameters. Subsequently, these parameters are extended to single particles of other shapes. Here, Φ=0.40, 0.60, 0.80 and 1.0 are taken as examples (but not limited to this), and their crushing properties are simulated and compared with the predictions of Eqs. (8) and (10).

5.1. Single-particle crushing process

In order to explain key processes such as single-particle crushing development and fracture evolution in detail, RB particles with a sphericity of Φ = 0.70 are selected as typical representatives in this section. The evolution process of particle crushing is shown in Figs. 15 and 16, and the corresponding stress-strain curve is shown in Fig. 17. This choice is primarily based on the following two reasons. First, the sphericity of Φ = 0.70 lies in the middle of the sphericity range (0.4–1.0) examined in this study, making it a better representative of the crushing behavior across different sphericities. Second, given that the simulation processes across all sphericities exhibit similar overall trends, selecting an instance with intermediate sphericity helps avoid lengthiness and repetition.

Figs. 15(a) and 16(a) show the particle just before the start of loading, in contact with the loading plate (ending the pre-loading state), with only a few fragments and noticeable force chains distributed at both ends of the loading plate and base. This corresponds to point a in the stress-strain curve in Fig. 17. As loading begins, initially, there is no significant change in the appearance and force state of the particle, as shown in Fig. 15(b) and point b in Fig. 17, though the number of fragments and force chains increases internally compared to the previous stage (Fig. 16(b)). Subsequently, cracks start to appear after a period of loading (Fig. 15(c)), and the first peak on the stress-strain curve is observed at point c in Fig. 17. At this stage, some fragments penetrate through, with force chains mainly concentrated around the fragments (Fig. 16(c)). After the first peak, the particle undergoes breaking, leading to the second (point d in Fig. 17) and third (point e in Fig. 17) peak points. The corresponding particle appearance and the distribution of internal fragments and force chains expand further (Fig. 16(d)(e) and 15 (d)(e)). The particle is essentially crushed upon reaching the system's maximum stress peak (point f in Fig. 17), with complete penetration of fragments and an expanded range of force chains (Fig. 16(f)), eventually breaking into 3–4 pieces, as shown in Fig. 15(g)-(j) and points g-j in Fig. 17. After peak f, the distribution of fragments and force chains expands from the penetrating central crack towards the right side (Fig. 16 (g)-(j)), indicating that after complete breaking, the largest fragment on the right side bears the main load, leading to further internal cracking.

Comparing the stress-strain curves of the single particle crushing test results (left side of Fig. 17) reveals that the numerical simulation results of this study basically match the experimental test results, with only a 1.05 % difference in the stress peak values. Additionally, the post-crushing crushing mode of the numerical simulation closely aligns with those observed in the experimental load crushing tests (Section 4.1, distribution of crushing), as shown on the right side of Fig. 17 (top view). This further demonstrates the feasibility of the numerical simulation method presented in this study.

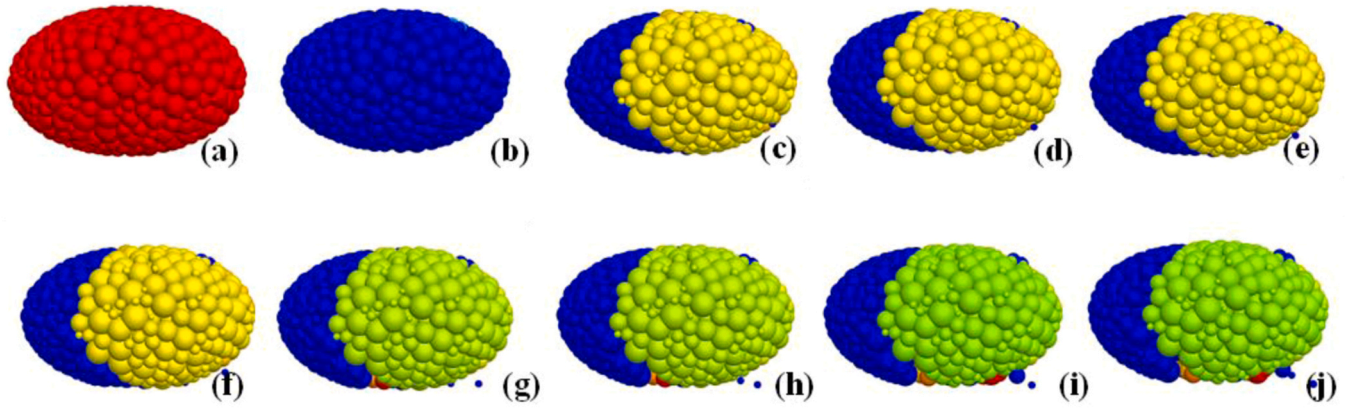


Fig. 15. Single-particle crushing process based on discrete element model.

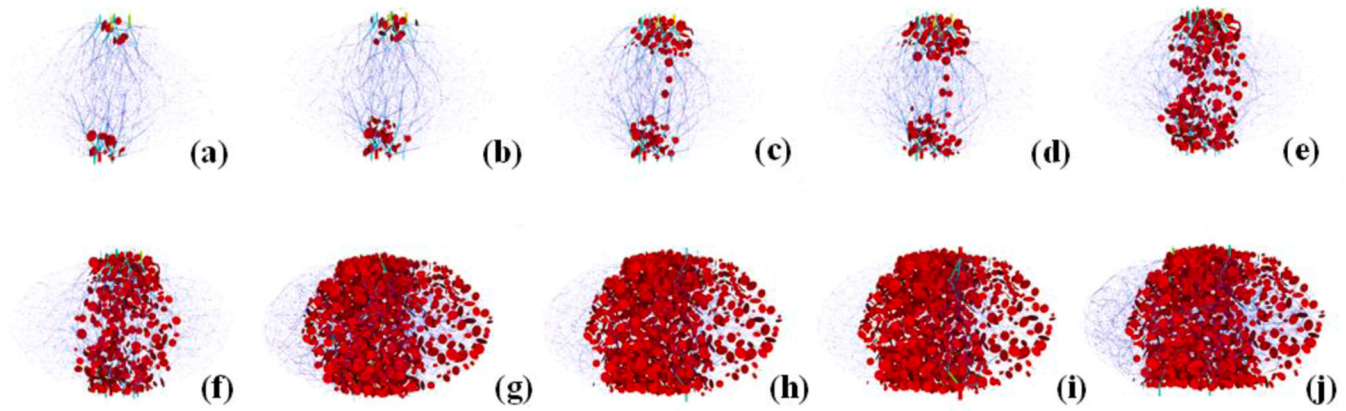


Fig. 16. Single-particle crushing force chain and fracture evolution process.

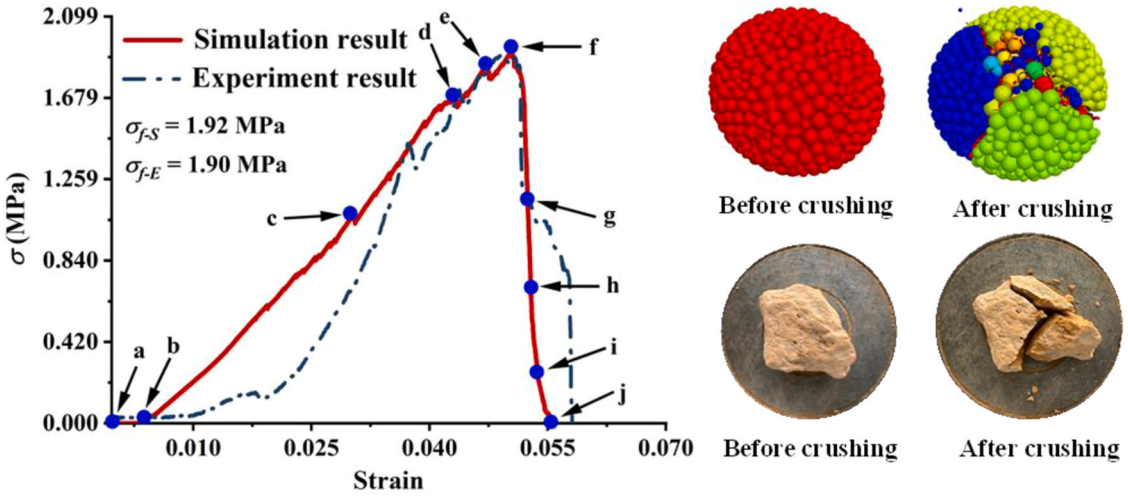


Fig. 17. Stress-strain curve of single-particle crushing based on discrete element model.

5.2. Numerical, experimental and predicted results comparison analysis

Utilizing PFC^{3D}, the average crushing strength and average crushing energy of single RA particles under the same conditions as the experimental test subjects (sphericities Φ=0.4–0.6, 0.6–0.8, and 0.8–1.0) were obtained. These results were compared with the outcomes of the experiments and the predictions made by Eqs. (8) and (10), as shown in Figs. 18 and 19.

Fig. 18 compares the results of the average crushing strength of single particles of RB, RC, and RM. From Fig. 18, it can be inferred that for single particles of different sphericities Φ, the experimental results, numerical results, and predictions from Eq. (8) are quite close to each other. Therefore, for the crushing strength of a single particle, the parameters determined are reasonable, and the numerical program can be used to simulate and calculate the crushing strength of particles with other sphericities. As shown in Fig. 19, which compares the average

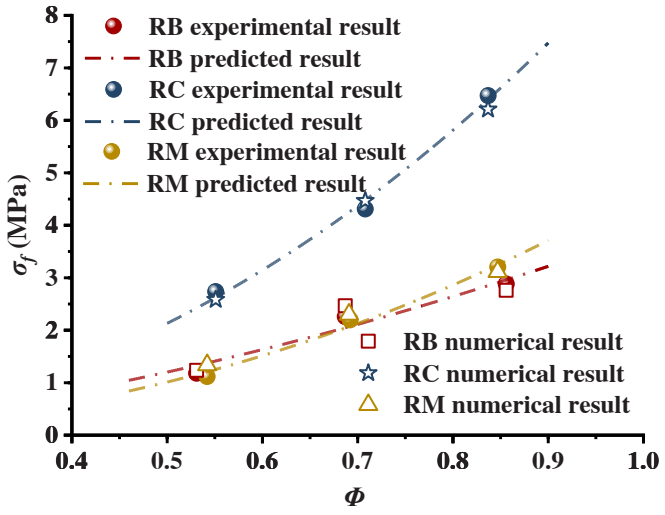


Fig. 18. Comparison of numerical, experimental and predicted results of single-particle average crushing strength.

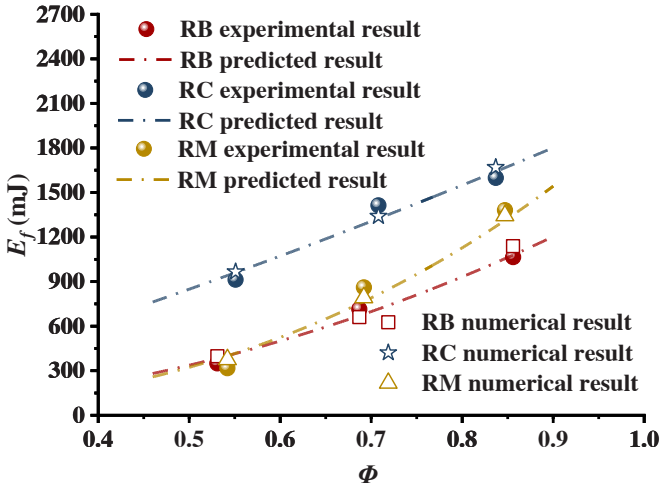


Fig. 19. Comparison of numerical, experimental and predicted results of single-particle average crushing energy.

crushing energy results of single particles of RB, RC, and RM, there are some differences among the average crushing energy experimental results, numerical results, and predictions made by Eq. (10) for the three different sphericities Φ. The error between the simulation results and the test results is between 2.66 % and 18.53 %, as shown in Table 3. The randomness of the experiments, as well as variations in shape, size, and mineral content among particles, lead to variations in the energy released during the experiments [17]. Nonetheless, overall, the differences in average crushing energy among single particles of different sphericities Φ are not pronounced. This indicates that although the results of individual experiments may vary due to various factors, considering a larger number of samples, shows that these results tend to be consistent, reflecting general stability in the relationship between crushing energy and particle sphericity Φ. Therefore, the numerical program can also be used to simulate and calculate the crushing energy of particles with other sphericities.

5.3. Discrete element simulation of single-particle crushing with different shapes and correction of prediction results

The study conducted simulations using the PFC^{3D} discrete element numerical program to calculate the crushing outcomes for single particles of RB, RC, and RM with sphericities Φ=0.40, 0.60, 0.80, and 1.0. The results are presented in Figs. 20 and 21. Fig. 20 compares the average crushing strength of single particles of RB, RC, and RM obtained from numerical results with those predicted by Eq. (8). It was found that, under specific sphericities Φ, the numerical results for average crushing strength are in good agreement with those obtained from Eq. (8). Under conditions of equivalent particle size, the greater the particle sphericity Φ, the greater the sustained crushing force, and correspondingly, the greater the crushing strength σ_f. Thus, the particle sphericity Φ and Eq. (8) can be used to predict the crushing strength of single particles with any sphericity, but the specific results may still require further experimental validation.

Additionally, Fig. 21 shows the trend in the average crushing energy of single particles of RB, RC, and RM. From Fig. 21(a), it can be seen that as the particle sphericity Φ increases, the crushing energy also increases, consistent with the conclusions of the aforementioned study. However, there are some discrepancies between the numerical results and the predictions made by Eq. (10). For particles with smaller sphericities (Φ<0.8), the numerical results are consistent with the predictions, but for larger sphericities ($\Phi\geq$0.8), Eq. (10) no longer fits well. The fitting correlation coefficient R^2 is generally low, and the fitting correlation coefficients R^2 of RB, RC and RM numerical results are 0.61, 0.54 and 0.77, respectively. Specifically, the predictions tend to overestimate compared to the simulation results (as indicated by the light purple area

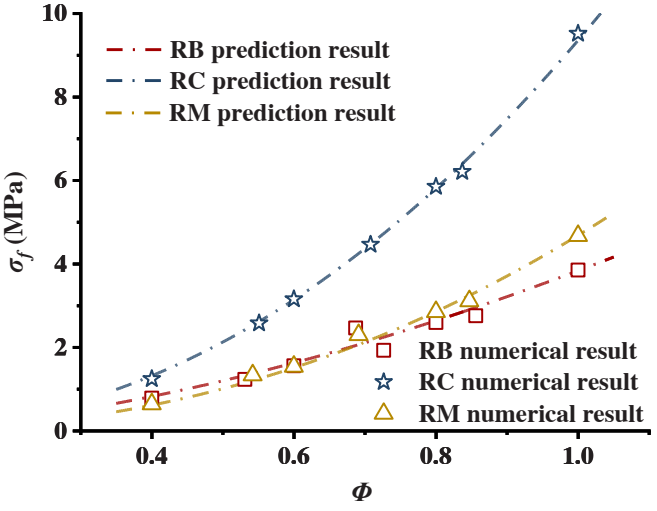


Fig. 20. Comparison of average crushing strength between numerical results and predicted results.

Table 3
Comparison of error between numerical results of single-particle crushing energy E_f of RA and experimental results.

Categories	Φ=0.4–0.6			Φ=0.6–0.8			Φ=0.8–1.0		
	RB	RC	RM	RB	RC	RM	RB	RC	RM
$E_{f\text{-}S}$ (mJ)	397.18	965.9	375.29	660.44	1340.47	792.04	1138.58	1669.36	1344.45
$E_{f\text{-}E}$ (mJ)	349.05	912.42	316.62	713.82	1413.27	860.55	1064.76	1598.63	1381.18
Error (%)	13.79	5.86	18.53	7.48	5.15	7.96	6.93	4.42	2.66

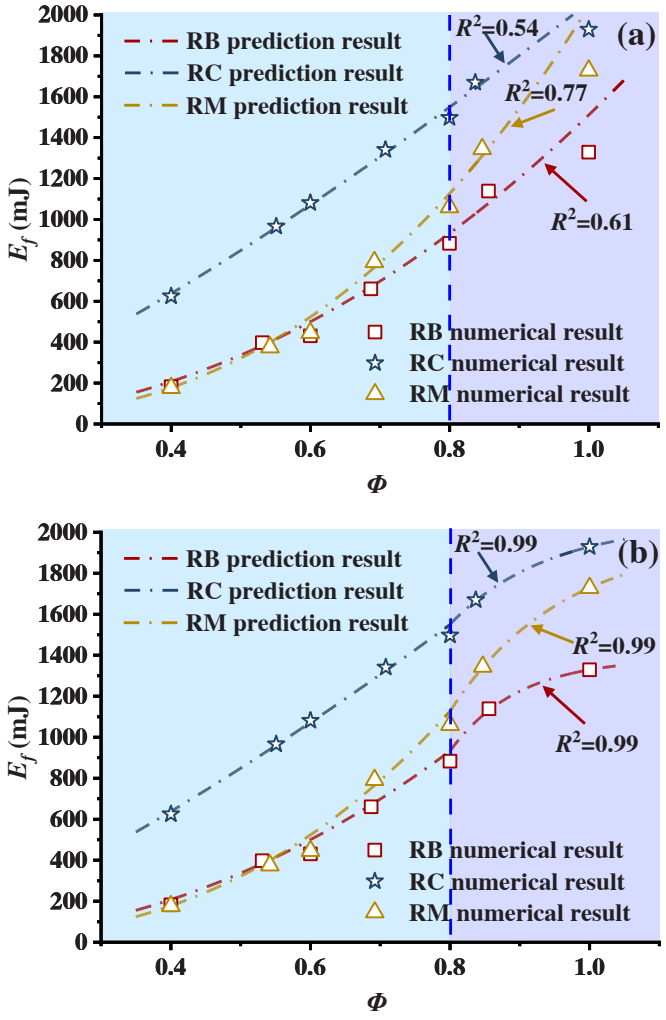


Fig. 21. Comparison of average crushing energy between numerical results and predicted results. (a) Comparison between numerical results and Eq. (10); (b) Comparison of numerical results with Eqs. (10) and (11).

in Fig. 21(a)), especially the predicted average crushing energy for RM with a sphericity Φ=1.0 by Eq. (10), which is almost equivalent to RC, not matching general perceptions, whereas the numerical results seem more reasonable. Therefore, based on numerical results, a correction of Eq. (10) is suggested, showing a certain pattern described by a new Eq. (11) as shown in Fig. 21(b). Eq. (11) involves nonlinear curve fitting parameters E_0, α, and t, with specific values for RB, RC, and RM provided in Table 4.

$$E_f = E_0 + 2\alpha \cdot \left(1 - e^{-\Phi/t}\right) \tag{11}$$

Hence, for the crushing energy of single particles with smaller sphericities (Φ<0.8), Eq. (10) can be used for predictions. However, for particles with larger sphericities ($\Phi \geq 0.8$) (as indicated by the light purple area in Fig. 21(b)), employing Eq. (11) for predicting their crushing energy is more appropriate. It is worth noting that since the nature of RA is heterogeneous due to different sources and compositions, this heterogeneity may limit the universality of the prediction equation based on a single material source to a certain extent. Of course, this single source of RA materials inevitably exists in many studies [2,6,9,35]. However, there is no doubt that this study provides some valuable preliminary insights into the crushing strength and crushing energy prediction equations of RA. Furtherly, this study regards these preliminary results as the cornerstone of further research, and will expand the data set (material source) and conduct rigorous statistical analysis in future research to further improve the broad applicability of the research results.

Table 4
Correction of prediction Eq. (11) fitting parameters.

Component categories	E_0	α	t
RB	-1.68×10^6	8.42×10^5	0.097
RC	-1.36×10^5	6.92×10^4	0.142
RM	-1.16×10^5	5.90×10^4	0.162

6. Conclusion and outlook

6.1. Research conclusion

This paper first quantitatively characterizes the shape characteristics of RA particles and then studies the crushing modes, fractal characteristics, particle crushing strength, and crushing energy of single particles with different shapes (sphericity Φ) through single-particle crushing experiments involving RB, RC, and RM. Finally, by comparing and analyzing the experimental and predicted results through PFC^{3D} discrete element numerical simulation, it predicts the crushing strength and crushing energy of particles with specific sphericity. The main conclusions are as follows.

(1) The shapes of RA particles are mainly disk, block, blade and bar. The flatness of the RA particles used in this study ranges from 0.30 to 0.96, the elongation ratio ranges from 0.29 to 0.98, the Domokos coefficient S_f ranges from 3.03 to 6.58, and the sphericity ranges from 0.42 to 0.93. The particle size of the RA has essentially no effect on their shape.

(2) The crushing modes of RB, RC, and RM particles can be divided into penetrating fractures, corner rupture, and overall fragmentation, with a correlation between the single-particle crushing mode and particle sphericity. The proportion of penetrating fractures and corner rupture tends to decrease with increasing sphericity, while the proportion of overall fragmentation tends to increase.

(3) Under different particle sphericities, the fractal dimensions of RB and RM are higher than those of RC, and the fractal dimensions of the three types of RA decrease with increasing particle sphericity, indicating a reduced degree of crushing. As the sphericity increases, the fractal characteristics of the RA weaken, and the gradation distribution narrows. The final fractal dimensions of multi-sphericity RB, RC, and RM are 2.47, 2.31, and 2.49, respectively.

(4) The particle crushing strength of RC under the same sphericity is higher than that of RB and RM. With the increase of particle sphericity, the crushing strength of RA shows an increasing trend, following an exponential growth model. Similarly, the relationship between particle crushing energy and particle sphericity also follows an exponential function, showing a clear effect of particle shape. The crushing strength and crushing energy of a single RA particle can be predicted by its sphericity.

(5) The results of the discrete element numerical simulation show that the single-particle crushing strength of RA can be calculated through sphericity and Eq. (8). For single-particle crushing energy with larger sphericities ($\Phi \geq 0.8$), Eq. (11) is superior to Eq. (10), thus its crushing energy can be calculated through sphericity and Eq. (11), and further validation through experiments is needed.

6.2. Research limitations and outlook

Additionally, while this study has yielded significant findings, there are still limitations that need to be addressed and several avenues for future research.

(1) Three materials are selected in this study, and there are some limitations in the types of materials. Future research should be extended to other types of construction waste to verify the universality of the conclusions.

(2) The current research does not fully cover all possible particle size distributions, and a wider range should be considered in the future to improve the applicability of the research results.

(3) This study mainly studied the influence of particle shape on the crushing characteristics, using a single loading rate and method, without considering the complex actual working conditions. In the future, more diversified loading conditions and environmental factors should be studied to fully understand the mechanical behavior of RA.

(4) This study mainly focuses on the initial crushing characteristics. In the future, the durability of recycled aggregate under long-term load and environmental action should be included to ensure its long-term stability in engineering.

In summary, addressing these limitations and expanding the research directions is expected to deepen the understanding of the crushing characteristics and mechanical properties of RA, thereby enhancing its potential for application in engineering practice.

CRediT authorship contribution statement

Peichen Cai: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation. **Xuesong Mao:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation, Conceptualization. **Qian Wu:** Writing – review & editing, Funding acquisition, Data curation. **Yueyue Wang:** Writing – original draft, Validation, Data curation.

Declaration of Competing Interest

We confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Data Availability

Data will be made available on request.

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